

Advanced Programming 2025

Can Sustainability Indicators Predict Economic Performance ?

Final Project Report

Gabriel Pinto Ferreira
gabriel.pintoferreira@unil.ch
Student ID: 22411383

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Abstract

This project investigates whether sustainability indicators, measured through Environmental, Social, and Governance (ESG) indices, are related to GDP growth and whether they can help predict it at the country level. Using data from the World Bank, ESG indicators are standardized and aggregated into the three composite indices and combined with economic variables for a sample of 50 countries. Two complementary approaches are applied. First, a fixed-effects regression is used to examine associations between ESG indices and annual GDP growth while controlling for country and year fixed effects. Then, supervised machine learning models are implemented to evaluate the predictive power of ESG indicators, comparing a linear regression baseline with two non-linear models. Results suggest that associations differ across ESG dimensions. Governance shows a positive coefficient, while Environmental and Social indices have negative or negligible associations. The machine learning analysis shows that non-linear models perform better than linear ones, but the overall predictive performance remains limited. These findings highlight the difficulty of predicting short-term economic growth using sustainability indicators alone.

Keywords: data science, Python, machine learning, sustainability, ESG, economic growth, fixed-effects regression, predictive performance

Contents

1	Introduction	3
2	Literature Review / Related Work	4
3	Methodology	5
3.1	Data Description	5
3.2	Approach	6
3.3	Implementation	8
4	Results	8
4.1	Experimental Setup	8
4.2	Performance Evaluation	9
4.3	Visualizations	10
5	Discussion	11
5.1	Fixed-effects regression results	11
5.2	Machine learning results	11
5.3	Differences in model performance	12
5.4	Challenges and limitations	12
5.5	Comparison with existing literature	12
6	Conclusion and Future Work	13
6.1	Summary	13
6.2	Future Directions	13
	References	14
A	Additional Tables and Figures	14
A.1	List of selected countries	14
A.2	Distribution of the sample of 50 countries	15
A.3	List of variables	16
A.4	Correlation heatmap of ESG and economic indicators	16
A.5	Full fixed-effects regression table	17
B	Code Repository	19
C	AI Usage Declaration	20

1 Introduction

Over the past decades, sustainability has become a global concern for countries' development and their economic policies. There are multiple ways to measure sustainability, one of which is through *Environmental, Social, and Governance (ESG)* indicators, which capture different dimensions, including environmental impact, social conditions, and the quality of institutions and governance. These indicators are commonly used to evaluate countries' and firms' sustainability performance.

Issues related to the three aspects of ESG indicators, such as atmospheric pollution, social inequality, and ethical governance, have pushed governments and international institutions to adapt their economic strategies to comply with sustainability concerns. However, despite the increasing consideration of ESG indicators in economic policies, their relationship with economic performance remains unclear. In this context, we can naturally ask the following question: does a relationship exist between sustainability indicators and the economic growth of countries, and can these indicators be used to predict it ?

The objective of this project is divided into two parts. First, the relationship between sustainability indicators and economic growth is examined through a fixed-effects regression. In this framework, the Environmental, Social, and Governance components of ESG are included separately, allowing the analysis to evaluate how each dimension relates to economic growth, while also controlling for unobserved country-specific characteristics that stay constant over time, as well as for common time effects affecting all countries.

The second objective is to evaluate the predictive power of sustainability indicators with respect to economic growth, using the same separation of the three ESG dimensions. To do so, three supervised machine learning models are tested: Linear Regression, Random Forest, and Gradient Boosting. By comparing the predictive performance of all three models, this approach complements the econometric analysis and provides insights into whether sustainability indicators contain useful information for forecasting economic growth.

It is important to state that the analyses conducted in this project do not aim to establish a causal relationship between sustainability indicators and economic growth. While the fixed-effects regression evaluates statistical associations and the machine learning models examine predictive performance, these methods do not allow for causal inference. The results should therefore be interpreted as descriptive and predictive, rather than causal.

The remainder of this report is structured as follows. Section 2 presents a review of the related literature and positions the project within the existing studies. Section 3 describes the data, technical approach, and implementation details. Section 4 presents the empirical results of the econometric and machine learning models. Section 5 provides a discussion of these results, and Section 6 concludes the report.

2 Literature Review / Related Work

Studying the relationship between sustainability and economic performance is not new to the literature. In fact, as sustainability has become more relevant in economic policy, many papers using ESG indicators to study how they are related to economic outcomes have been published¹. However, these studies often differ in their empirical methods, choice of outcome variables, and the way ESG indicators are constructed, which leads to mixed results.

One recent study closely related to this project is Wang et al. (2023), who examine the relationship between country-level ESG indicators and economic growth. Similarly to this project, the authors explicitly separate the Environmental, Social, and Governance components of ESG and analyze their effects on GDP growth. Using a panel of 109 countries over the period 1996-2019, combining country-level ESG indicators and macroeconomic data, their empirical approach relies on panel regressions with fixed effects and additional strategies aimed at identifying causal relationships. The results of this paper suggest that governance-related indicators are positively associated with economic growth, while environmental and social indicators have more heterogeneous effects. While this study provides valuable evidence on the role of ESG in economic growth, the analysis is mainly causal, and does not provide any insight whether sustainability is a good predictor for economic performance.

Another related study is Diaye et al. (2022), who analyze the relationship between sustainability and economic performance using GDP per capita as the main outcome variable. Their analysis relies on a panel dataset from 1996-2014 including the 29 OECD countries. Using cointegration methodology, the authors find that sustainability indicators are positively associated with GDP per capita in the long-term, while short-term effects are more limited. While these results are important, the focus of this study is primarily on long-term relationships, whereas this project concentrates more on the short-term by using GDP growth as the dependent variable.

Lastly, another important contribution is Bota-Avram et al. (2018), who examine the relationship between governance quality and economic performance at the country level. Using data from the World Bank, this paper combines panel regression with Granger non-causality tests to study how governance-related factors are associated with economic growth and other macroeconomic metrics. The results suggest that better governance is positively related to economic performance. Although this study only relies on one aspect of ESG, it highlights the potential value of analyzing ESG dimensions separately.

Overall, the existing literature provides important evidence of the relationship between sustainability and economic performance at the country level, but it also leaves several aspects relatively unexplored. This project contributes to the literature by offering a complementary, non-causal analysis of the relationship between sustainability and economic performance in the short-term, while also providing additional evidence using machine learning models to measure the predictive power of ESG indicators with respect to GDP growth.

¹ All papers mentioned in this section are included in the References.

3 Methodology

3.1 Data Description

The data used for this project is constructed from publicly available sources provided by the World Bank ². The raw data is obtained from three separate datasets. The first one contains a set of ESG indicators covering 192 countries over the period 2000-2023. The second dataset includes a range of macroeconomic indicators for the same countries and time period, including GDP growth. Finally, the third dataset provides country-level characteristics, including regional classification and income level (treated as static over the sample period).

These three datasets were then merged at the country-year level to construct a unified panel dataset. The resulting panel reports the values of 16 indicators for each country and year, combining ESG indicators and economic variables.

From this merged dataset, a sample of 50 countries was selected for the future analysis ³. The sample selection was based on three criteria: geographic region, income level, and data availability. Countries were chosen to ensure representation across all different regions and income groups, while also limiting the impact of missing observations ⁴. For that matter, each country included in the final sample has at the very least 70% of available data. The resulting panel of 50 countries contains 19'200 observations ⁵.

The variables in the dataset play distinct roles in the analysis. Sustainability indicators are used to capture the Environmental, Social, and Governance dimensions of countries' development. The macroeconomic variables are included to measure economic performance, with GDP growth serving as the main outcome variable, while additional economic indicators are used exclusively in the machine learning models as predictive features. Country-level characteristics, such as region and income level, are not directly used in the models but are taken into account for the sample construction and descriptive purposes.

Data Quality represents an important challenge in this project, as the availability of ESG and economic indicators can vary substantially. While most of the indicators have a relatively satisfying data coverage, there are several others, especially those related to social conditions, that contain a non-negligible proportion of missing observations.

Figure 1 illustrates the percentage of missing values for each indicator included in the analysis. The figure shows a heterogeneous pattern across variables, with most of the social indicators having close to half their data missing, while most of the economic indicators having nearly no missing values at all. This uneven data availability results in an unbalanced panel, as not all indicators are observed for each country-years observation.

To limit the impact of missing data on the analysis, two measures were taken. First, as mentioned earlier, the sample of 50 countries is restricted to those with at least 70% of data available. Second, the analysis is conducted using only available data. No imputation techniques were applied, avoiding the results representing generated data instead of actual reported values.

Despite these precautions, missing data still remains one of the main limitations of this project, and may reduce the effective sample size for certain models, particularly machine learning, where we need to have a clean dataset, with no missing values.

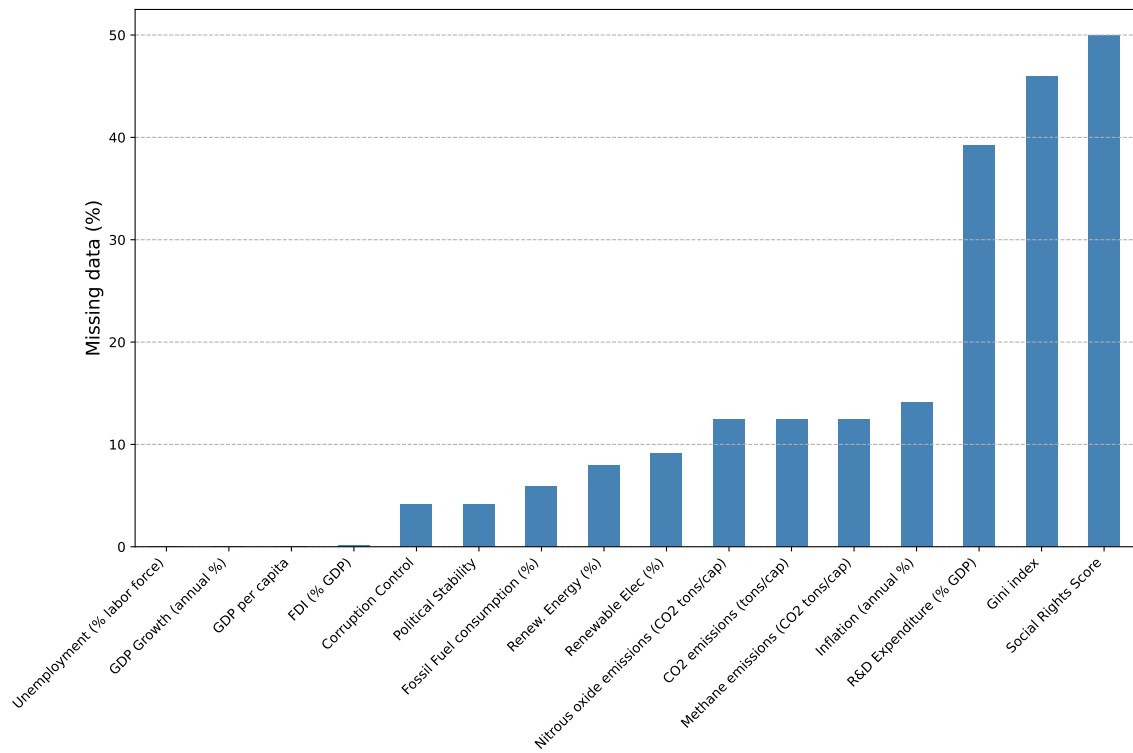
² All data sources are listed in the References section.

³ A list of the 50 selected countries is provided in the Additional Tables and Figures section.

⁴ The distribution of the 50 selected countries across regions and income groups is reported in the additional Tables and Figures section.

⁵ To clarify the composition of the dataset, a table summarizing the 16 indicators and their corresponding ESG or economic categories is provided in the Additional Tables and Figures Section.

Figure 1: Percentage of missing values by indicator



3.2 Approach

This project applies two complementary methodological approaches, each one serving a distinct objective. First, a fixed-effects panel regression is used to analyze the relationship between sustainability indicators and GDP growth at the country level.

The model specification is written as follows:

$$\text{GDPGrowth}_{i,t} = \beta_0 + \beta_1 \text{ENV}_{i,t} + \beta_2 \text{SOC}_{i,t} + \beta_3 \text{GOV}_{i,t} + \gamma_i + \delta_t + \varepsilon_{i,t}$$

where:

- $\text{GDPGrowth}_{i,t}$ denotes the annual GDP growth rate of country i in year t .
- $\text{ENV}_{i,t}$, $\text{SOC}_{i,t}$, and $\text{GOV}_{i,t}$ represent the Environmental, Social, and Governance indices for country i at time t , respectively.
- β_0 is the constant, while β_1 , β_2 , and β_3 capture the association between each ESG dimension and economic growth.
- γ_i denotes country fixed effects, controlling for time-invariant country-specific characteristics.
- δ_t denotes year fixed effects, capturing common shocks affecting all countries in a given year.
- $\varepsilon_{i,t}$ is the error term.

In this framework, the Environmental, Social, and Governance components of ESG are included separately, allowing each dimension to be analyzed individually. By including both country and year fixed effects, the regression focuses on within-country variation over time. This setting is suitable to examine how changes in one of the ESG dimensions for a given country at a certain point in time are associated with variations in GDP growth.

The second approach focuses on the predictive power of sustainability indicators with respect to economic growth. In this part of the analysis, supervised machine learning models are employed to evaluate whether ESG indicators contain useful information for forecasting GDP growth at the country level.

Three models are considered:

- **Linear Regression:**

A baseline model used to evaluate predictive performance under linear assumptions that also provides a direct comparison with the econometric approach.

- **Random Forest:**

A tree-based model that combines the predictions of many decision trees built on different subsets of the data, allowing it to capture non-linear relationships.

- **Gradient Boosting:**

A second tree-based model that builds decision trees sequentially, enabling the model to learn more complex patterns and improve prediction accuracy.

Before applying these models, several data processing steps are required to ensure the dataset is prepared for the analysis.

First, the ESG indices are constructed by standardizing each ESG indicator using the z-score formula :

$$z_{i,t} = \frac{x_{i,t} - \mu_x}{\sigma_x}$$

where $x_{i,t}$ is the original value of the indicator for country i at time t , μ_x and σ_x are the mean and standard deviation of the indicator respectively, computed using all country-year observations for the corresponding indicator. The direction of each indicator was adjusted so that an increase in an indicator means a better ESG performance. The standardized indicators are then grouped by ESG category and aggregated together to form indices. Each index is built at the country-year level and is intended to summarize the overall performance of a country at a certain time for one of the three ESG dimensions. It also helps to limit the influence of strong correlations among closely related indicators ⁶, such as emissions-related measures.

Next, missing values are handled differently depending on the approach. For the fixed-effects regression, the missing values were kept. No imputation was applied, and no part of the dataset was removed. This choice avoids introducing artificial information into the regression results. As for the machine learning models, a different approach was required. In fact, only observations for which all required features and the target variable are available were kept in the dataset, in order for the models to work properly.

Then, feature scaling is applied where necessary. For the machine learning analysis, feature scaling is applied to the Linear Regression model, as this model is sensitive to the scale of its features. On the other hand, no additional scaling is required for the two tree-based models.

⁶ A correlation heatmap illustrating pairwise correlations between the indicators is provided in the Additional Tables and Figures section.

Finally, the dataset used for machine learning is split into training and test sets, using a 80/20 split. Cross-validation is performed exclusively on the training set to evaluate the models robustness, while the test set is kept aside to measure the out-of-sample predictive performance.

To evaluate the performance of both the fixed-effects regression and machine learning models, several evaluation metrics are used. The first is the coefficient of determination (R^2). It measures the proportion of the variance in GDP growth that is explained by the model and provides an indication of its explanatory power. This metric is used for all econometric and machine learning models.

Along with the R^2 , the Root Mean Square Error (RMSE) is used to measure prediction accuracy. RMSE represents the average difference between the predicted and observed values of GDP growth. Because it is expressed in the same unit as GDP growth, it provides an intuitive measure of prediction error.

Lastly, cross-validation is used to obtain a more reliable measure of predictive performance. Instead of evaluating the models with a single train/test split, the training data is divided into several subsets. The model is then trained and evaluated multiple times, each time using a different subset. The final cross-validation score corresponds to the average performance of all these repetitions.

3.3 Implementation

This project is implemented entirely in Python. It is used for data collection and preparation, econometric analysis, and machine learning modeling. Data manipulation and preparation are performed using the pandas and numpy libraries. The fixed-effects regression is estimated using the statsmodels package, which provides flexible tools for econometric modeling. The machine learning models are implemented using the scikit-learn library, which also provides utilities for train/test split and cross-validation. Finally, figures are generated using matplotlib.

The overall workflow is organized using a modular architecture. A main script (main.py) serves as the entry point and sequentially executes each stage of the analysis. Data preparation and cleaning are handled in a dedicated module, followed by country selection and construction of the final sample of countries. The econometric analysis and machine learning models are implemented in separate modules, each producing their own outputs. Intermediate datasets and final results are saved to the repository, including processed data files, regression tables, and figures, allowing each step of the pipeline to be inspected independently.

4 Results

4.1 Experimental Setup

All analyses are conducted using Python (version 3.12.11). The code uses basic libraries. Data manipulation is performed using pandas and numpy. The fixed-effects regression is estimated using the statsmodels package, while machine learning models are implemented using scikit-learn. Additionally, figures are generated using matplotlib.

For the fixed-effects regression, standard errors are clustered at the country level to reflect that shocks affecting a country may persist across the years.

Regarding the machine learning models, hyperparameters for the two non-linear models (Random Forest and Gradient Boosting) are set in a way that reduces overfitting. For instance, the Random Forest model is estimated using 300 trees, a maximum depth of 6, and a minimum of 10 observations per leaf. The Gradient Boosting model uses 800 trees, a learning rate of 0.03, maximum tree depth of 2, a minimum of 20 observations per leaf, and a subsampling rate of 0.7.

4.2 Performance Evaluation

The first set of results reports the estimates from the fixed-effects regression, which examines the relationship between sustainability indicators and GDP growth at the country level. The dependent variable is GDP growth, while the environmental, social, and governance indices are included as explanatory variables. Country and year fixed effects are included in the specification to control for time-invariant country characteristics and common shocks across years, but they are not shown in the following table for better readability ⁷.

Table 1: Fixed-effects regression results

Dep. Variable:	gdp_growth	R-squared:	0.516
No. Observations:	1117	Adj. R-squared:	0.482
F-statistic:	25.17	Prob (F-statistic):	4.21e-20
Covariance Type:	cluster		

	coef	std err	z	P > z	[0.025	0.975]
Intercept	1.4058	1.979	0.711	0.477	-2.472	5.284
ENV_index	-0.6877	0.812	-0.847	0.397	-2.280	0.904
SOC_index	-0.0427	0.191	-0.224	0.823	-0.416	0.331
GOV_index	0.3860	0.356	1.085	0.278	-0.311	1.083

Notes:

[1] Standard Errors are robust to cluster correlation (cluster)

The reported coefficients in the table describe how changes in sustainability indicators within a given country over time are associated with changes in GDP growth. Because the model includes country and year fixed effects, the estimates only rely on within-country variations and ignore differences that are constant over time or common to all countries. Standard errors are clustered at the country level to account for the fact that observations from the same country may be correlated across years. The table also reports the R-squared and adjusted R-squared as measures of the overall model fit. The column $P > |z|$ reports the statistical significance of each coefficient and shows whether the estimated association between an ESG dimension and GDP growth is statistically different from zero.

While the fixed-effects regression focuses on statistical associations, the machine learning models are evaluated based on their predictive performance. To allow for a direct comparison between the three models, the following table shows a comparison between key performance metrics for each model, including the R^2 , the RMSE, and cross-validation scores.

Table 2: Machine learning model performance comparison

Model	CV R^2	CV RMSE	Test R^2	Test RMSE
Linear Regression	0.022	3.464	0.143	3.137
Random Forest	0.166	3.200	0.327	2.781
Gradient Boosting	0.103	3.318	0.358	2.715

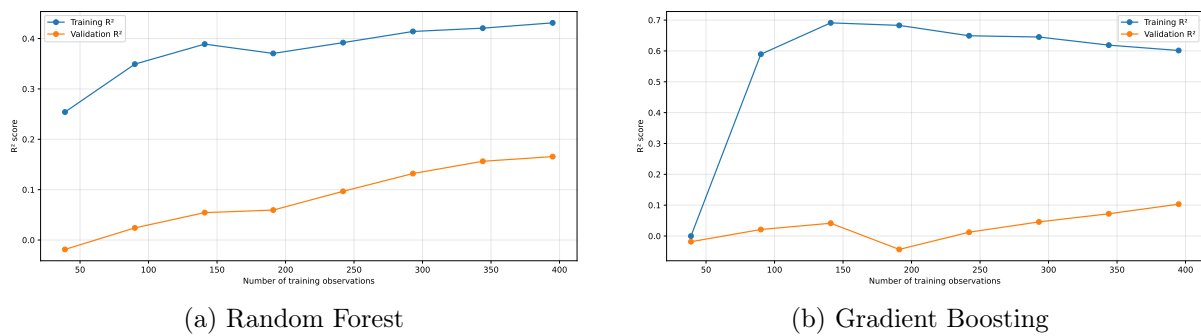
⁷ The complete regression table including the country and year fixed effects is shown in the Additional Tables and Figures section.

4.3 Visualizations

To complement the numerical results reported previously, this subsection presents a set of figures that help understand the behavior of the machine learning models.

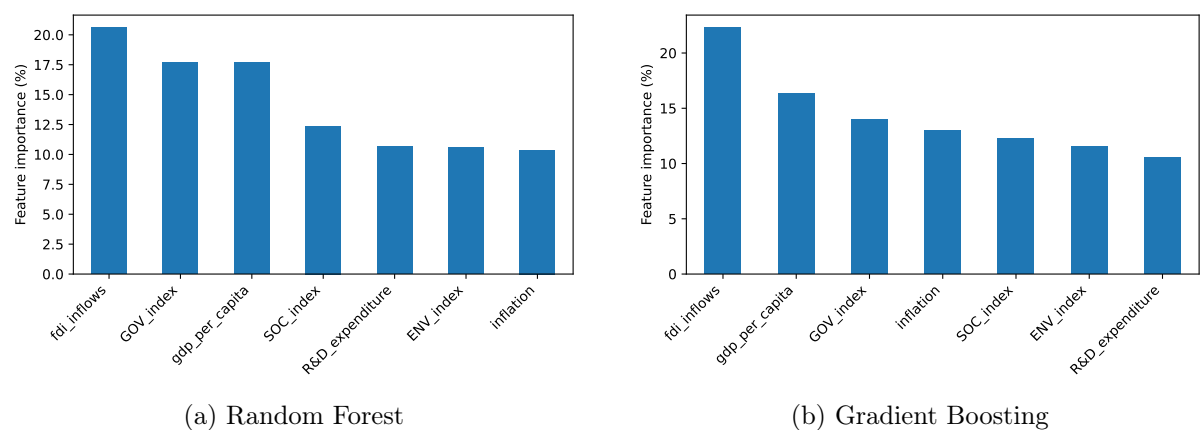
First, learning curves are used to visualize how model performance evolves as the size of the training dataset increases. For both Random Forest and Gradient Boosting models, the learning curves report training and cross-validation performance (using the R^2) as a function of the number of training observations. These plots help identify the presence of overfitting or underfitting, and indicate whether increasing the size of the training dataset could improve model performance.

Figure 2: Learning curves for the tree-based machine learning models



Along with the learning curves, feature importance plots are also used to examine how each feature contributes to the prediction of GDP growth. These plots display the percentage contribution of the ESG indices and the economic variables included as features in the Random Forest and Gradient Boosting models.

Figure 3: Feature importance for the Random Forest and Gradient Boosting models



5 Discussion

5.1 Fixed-effects regression results

As we can see from the fixed-effects regression results shown in Table 1, the estimated coefficients differ across the three ESG dimensions, on both sign and magnitude. The Environmental index is associated with a negative coefficient. The value of the coefficient implies that a one-unit increase in the Environmental index (i.e. a one-standard deviation increase) is associated with a decrease in GDP growth of 0.6877 percentage points. On the other hand, the Governance index shows a positive association with GDP growth. As for the Social index, it is close to zero, meaning it has practically no association with the dependent variable.

The R^2 of this model is 0.516, meaning that approximately 51.6% of the variance is explained by the model, which indicates a moderate explanatory power. However, none of these coefficients are statistically significant, as indicated by their p-values. This suggests that, in this specification, changes in ESG indices are not strongly linked to short-term variations in GDP growth.

Several factors may explain these results. First, GDP growth is a short-term and highly volatile measure that depends on many economic factors not captured by ESG indicators. In addition, improvements in ESG performance may affect economic outcomes mainly in the long-term, rather than through immediate changes in annual rates. Data limitations, especially the high percentage of missing values for some Social indicators, may also have weakened the estimated relationships.

Overall, the fixed-effects regression suggests that there is limited evidence of a clear association between ESG performance and economic growth.

5.2 Machine learning results

As shown by Table 2, the baseline linear regression model performs poorly, with low values in both cross-validated and test R^2 . This result indicates that a simple linear relationship between ESG indices and GDP growth isn't sufficient to capture patterns in the data. This deduction is consistent with the conclusion made for the fixed-effects regression discussed before.

In contrast, the two non-linear models achieve noticeably higher predictive performance. Both Random Forest and Gradient Boosting models obtain higher test R^2 values and lower test RMSE compared to the linear regression model, indicating a better ability to predict GDP growth out of sample. Among these two tree-based models, Gradient Boosting slightly outperforms Random Forest with a test R^2 value of 0.358 and RMSE of 2.715. However, even when performing better than Linear Regression, the non-linear models' performance still remains modest.

The learning curves shown in Figure 2 help explain these results. For both models, the training R^2 is substantially higher than the cross-validation R^2 , especially for smaller sample sizes. These gaps point to a degree of overfitting. The two learning curves also show that increasing the number of training observations reduces the gaps between training and cross-validation R^2 , suggesting that additional data could help stabilize predictions and reduce overfitting.

Figure 3 further illustrates how different variables contribute to the predictions of GDP growth. In both non-linear models, economic variables such as foreign direct investment and GDP per capita appear among the most influential features of the models. For the ESG indices, only the Governance index plays a substantial role. This indicates that, while ESG indicators contribute to the prediction, they are not the main predictive factors.

To conclude, the machine learning results show that non-linear models are better suited for predicting GDP growth. However, the presence of overfitting and modest predictive performance highlight the challenge of forecasting economic variables.

5.3 Differences in model performance

The differences in performance across the machine learning models mainly come from their ability to capture non-linear patterns in the data. The linear regression model assumes a simple linear relationship between ESG indices and GDP growth, which is not the case in reality. In fact, economic growth is driven by many factors, and the effect of sustainability is unlikely to be linear or constant over time.

On the other hand, the tree-based models (Random Forest and Gradient Boosting) can capture non-linear relationships and interactions between variables, explaining their higher predictive performance. Among them, Gradient Boosting is the one that slightly outperforms the other, but it's also the one that shows stronger signs of overfitting, as indicated by the gap between the training and cross-validation R^2 . Overall, the more flexible model (Gradient Boosting) wins, but the gain is very limited.

5.4 Challenges and limitations

One important limitation of this project is the size of the final dataset. Although the 50-country sample dataset contains a large number of observations, the quantity of data drastically decreases when removing all missing values for the machine learning models. This reduction limits the predictive power of the models, that would perhaps have had better results if more data was available.

Another challenge comes from the nature of the dependent/target variable. Annual GDP growth is a highly volatile variable and is influenced by many factors. For the fixed-effects regression, this volatility can make it harder to identify potential relationships with sustainability indicators, especially since ESG-related effects generally affect the long-term more.

The machine learning analysis also faces limitations related to overfitting. Despite the use of cross-validation and specific hyperparameters, the learning curves still reveal a persistent gap between the training and validation performance for both non-linear models. This suggests that the models may be capturing noise or sample-specific patterns rather than predictive relationships.

Finally, the construction of the ESG indices itself introduced limitations. Aggregating multiple standardized indicators into composite indices inevitably leads to some loss of information and assumes all indicators have equal weight. While this approach simplifies the analysis and reduces multicollinearity, it also reduces its precision.

5.5 Comparison with existing literature

The results of this project are largely consistent with the findings of previous related work. In the fixed-effects regression, we could see that the Governance index is the only dimension of ESG showing a positive association with GDP growth. This aligns with the findings of Wang et al. (2023), who state that governance-related indicators are positively associated with economic growth, as well as with Bota-Avram et al. (2018), who highlight the important role of governance quality in explaining the economic performance of a country. Although this estimated coefficient in this project is not statistically significant, its sign and magnitude are in line with these earlier studies.

At the same time, the relatively weak results found for the fixed-effects regression and machine learning models are consistent with the study of Diaye et al. (2022), who find that sustainability indicators are mainly associated with economic performance in the long-term, while short-term effects are weaker. Since this project focuses on annual GDP growth, which is highly volatile, it is not surprising that both explanatory and predictive power remain modest.

6 Conclusion and Future Work

6.1 Summary

To conclude, this project had the objective of examining whether sustainability indicators, measured through Environmental, Social, and Governance (ESG) indices, are related to and can help predict economic growth at the country level. To address this question, the analysis combined a fixed-effects regression with supervised machine learning models, focusing on annual GDP growth as the main output variable.

The econometric results suggest that the relationship between ESG performance and short-term economic growth is limited. Among the three ESG dimensions, the Governance index shows a positive association with GDP growth, while the Environmental and Social indices have negative or negligible relationships. Additionally, none of the estimated coefficients are statistically significant, meaning that changes in ESG indices aren't strongly linked to variations in GDP growth.

The machine learning analysis complements these findings from a predictive perspective. The Linear Regression model performance is weak, suggesting that linear relationships aren't sufficient to capture patterns in the data. The two non-linear models on the other side achieve better predictive performance, although accuracy remains limited, with both models suffering from some degree of overfitting. The feature importance plots reveal that only the Governance index plays a substantial role in the performance of the models, which coincides with the related work in the literature.

Overall, the results indicate that ESG indices alone have very limited explanatory and predictive power for economic growth. Governance indicators seem to be more relevant than the other sustainability dimensions prediction-wise, but predicting GDP growth remains very challenging due to its natural volatility and the influence of many factors beyond ESG indicators.

6.2 Future Directions

There are several extensions to this project that could be considered to improve the analysis.

The first consideration is the time horizon of the analysis. Since ESG indicators are more likely to affect economic outcomes in the long-run, future work could focus on long-term measures and use a dataset with a longer time span to better capture the economic effects of sustainability-related factors.

The construction of the ESG indices could also be refined. In this project, the indices are built by combining several standardized indicators into a single measure of ESG dimensions. While this approach makes the results easier to interpret, it inevitably leads to some loss of information. A future study could explore alternative ways of measuring ESG performance, for example by constructing the ESG indices with a different method (perhaps by giving different weights to the indicators) or by avoiding composite indices altogether and analyze each indicator separately.

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A Additional Tables and Figures

A.1 List of selected countries

Table 3: List of the 50 selected countries for the sample

Country	Region	Income level
Algeria	Middle East, North Africa, Afghanistan & Pakistan	Upper middle income
Australia	East Asia & Pacific	High income
Bangladesh	South Asia	Lower middle income
Bhutan	South Asia	Lower middle income
Bolivia	Latin America & Caribbean	Lower middle income
Brazil	Latin America & Caribbean	Upper middle income
Burkina Faso	Sub-Saharan Africa	Low income
Cambodia	East Asia & Pacific	Lower middle income
Canada	North America	High income
Chile	Latin America & Caribbean	High income
China	East Asia & Pacific	Upper middle income
Colombia	Latin America & Caribbean	Upper middle income
Costa Rica	Latin America & Caribbean	High income
Dominican Rep.	Latin America & Caribbean	Upper middle income
Egypt, Arab Rep.	Middle East, North Africa, Afghanistan & Pakistan	Lower middle income
France	Europe & Central Asia	High income
Georgia	Europe & Central Asia	Upper middle income
Germany	Europe & Central Asia	High income

Continued on next page

Country	Region	Income level
Ghana	Sub-Saharan Africa	Lower middle income
Honduras	Latin America & Caribbean	Lower middle income
Hungary	Europe & Central Asia	High income
India	South Asia	Lower middle income
Indonesia	East Asia & Pacific	Upper middle income
Iran, Islam Rep.	Middle East, North Africa, Afghanistan & Pakistan	Upper middle income
Israel	Middle East, North Africa, Afghanistan & Pakistan	High income
Japan	East Asia & Pacific	High income
Kazakhstan	Europe & Centra Asia	Upper middle income
Kenya	Sub-Saharan Africa	Lower middle income
Korea, Rep.	East Asia & Pacific	High income
Madagascar	Sub-Saharan Africa	Low income
Maldives	South Asia	Upper middle income
Mauritius	Sub-Saharan Africa	Upper middle income
Mexico	Latin America & Caribbean	Upper middle income
Nepal	South Asia	Lower middle income
Nigeria	Sub-Saharan Africa	Lower middle income
Pakistan	Middle East, North Africa, Afghanistan & Pakistan	Lower middle income
Peru	Latin America & Caribbean	Upper middle income
Philippines	East Asia & Pacific	Lower middle income
Poland	Europe & Central Asia	High income
Romania	Europe & Central Asia	High income
Rwanda	Sub-Saharan Africa	Low income
Saudi Arabia	Middle East, North Africa, Afghanistan & Pakistan	High income
South Africa	Sub-Saharan Africa	Upper middle income
Sri Lanka	South Asia	Lower middle income
Tunisia	Middle East, North Africa, Afghanistan & Pakistan	Lower middle income
United Kingdom	Europe & Centra Asia	High income
United States	North America	High income
Uruguay	Latin America & Caribbean	High income
Uzbekistan	Europe & Central Asia	Lower middle income
Vietnam	East Asia & Pacific	Lower middle income

A.2 Distribution of the sample of 50 countries

Table 4: Distribution of selected countries by region and income level

	High income	Low income	Lower middle	Upper middle	Total
East Asia & Pacific	3	0	3	2	8
Europe & Central Asia	6	0	1	2	9
Latin America & Caribbean	3	0	2	5	10
Middle East & North Africa	2	0	3	2	7
North America	2	0	0	0	2
South Asia	0	0	5	1	6
Sub-Saharan Africa	0	3	3	2	8
Total	16	3	17	14	50

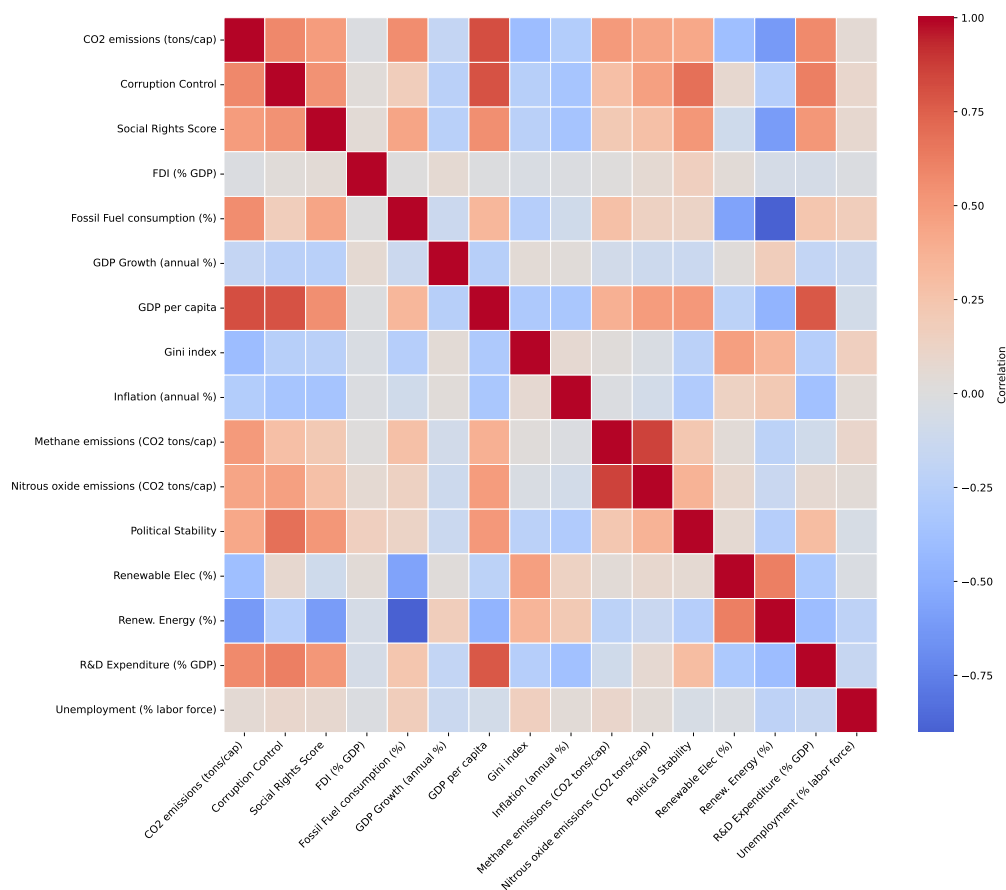
A.3 List of variables

Table 5: List of the ESG and economic variables

Variable	Category
CO2 emissions (metric tons per capita)	Environmental
Methane emissions (metric tons of CO2 equivalent per capita)	Environmental
Nitrous oxide emissions (metric tons of CO2 equivalent per capita)	Environmental
Fossil fuel energy consumption (% of total)	Environmental
Renewable energy consumption (% of total final energy consumption)	Environmental
Renewable electricity output (% of total electricity output)	Environmental
Unemployment, total (% of total labor force)	Social
Gini index	Social
Economic and Social Rights Performance Score	Social
Control of Corruption: Estimate	Governance
Political Stability and Absence of Violence/Terrorism: Estimate	Governance
GDP growth (annual %)	Economic
GDP per capita (constant 2015 US\$)	Economic
Inflation, consumer prices (annual %)	Economic
Research and development expenditure (% of GDP)	Economic
Foreign direct investment, net inflows (% of GDP)	Economic

A.4 Correlation heatmap of ESG and economic indicators

Figure 4: Correlation heatmap of ESG and economic indicators



A.5 Full fixed-effects regression table

Table 6: Complete fixed-effects regression table

Dep. Variable:	gdp_growth	R-squared:	0.516
Model:	OLS	Adj. R-squared:	0.482
Method:	Least Squares	F-statistic:	25.17
Prob (F-statistic):	4.21e-20	Log-Likelihood:	-2704.6
No. Observations:	1117	AIC:	5559.
Df Residuals:	1042	BIC:	5935.
Df Model:	74	Covariance Type:	cluster

	coef	std err	z	P > z	[0.025]	[0.975]
Intercept	1.4058	1.979	0.711	0.477	-2.472	5.284
C(country_code)[T.BFA]	5.1807	2.521	2.055	0.040	0.239	10.122
C(country_code)[T.BGD]	5.8841	2.191	2.686	0.007	1.590	10.178
C(country_code)[T.BOL]	2.9969	1.778	1.685	0.092	-0.488	6.482
C(country_code)[T.BRA]	1.9151	2.204	0.869	0.385	-2.405	6.235
C(country_code)[T.BTN]	5.4093	2.836	1.908	0.056	-0.149	10.967
C(country_code)[T.CAN]	0.2480	1.233	0.201	0.841	-2.169	2.664
C(country_code)[T.CHL]	2.4155	1.887	1.280	0.200	-1.282	6.113
C(country_code)[T.CHN]	7.6482	1.748	4.375	0.000	4.222	11.075
C(country_code)[T.COL]	3.4328	2.173	1.580	0.114	-0.826	7.692
C(country_code)[T.CRI]	3.3336	2.358	1.413	0.158	-1.289	7.956
C(country_code)[T.DEU]	-0.4091	1.484	-0.276	0.783	-3.317	2.499
C(country_code)[T.DOM]	4.0978	1.790	2.289	0.022	0.589	7.607
C(country_code)[T.DZA]	2.0497	1.580	1.297	0.195	-1.047	5.146
C(country_code)[T.EGY]	3.7655	1.809	2.081	0.037	0.219	7.312
C(country_code)[T.FRA]	0.1588	1.751	0.091	0.928	-3.274	3.591
C(country_code)[T.GBR]	0.1175	1.436	0.082	0.935	-2.697	2.932
C(country_code)[T.GEO]	5.0983	2.167	2.353	0.019	0.851	9.345
C(country_code)[T.GHA]	5.4607	2.453	2.226	0.026	0.652	10.269
C(country_code)[T.HND]	3.7498	2.381	1.575	0.115	-0.916	8.416
C(country_code)[T.HUN]	1.2561	1.616	0.777	0.437	-1.912	4.424
C(country_code)[T.IDN]	4.4633	2.010	2.220	0.026	0.523	8.404
C(country_code)[T.IND]	5.7811	2.122	2.724	0.006	1.622	9.941
C(country_code)[T.IRN]	2.2235	1.519	1.464	0.143	-0.753	5.200
C(country_code)[T.ISR]	2.7932	1.414	1.975	0.048	0.022	5.565
C(country_code)[T.JPN]	-0.7922	1.473	-0.538	0.591	-3.680	2.095
C(country_code)[T.KAZ]	4.1736	1.187	3.516	0.000	1.847	6.500
C(country_code)[T.KEN]	4.8289	2.934	1.646	0.100	-0.922	10.580
C(country_code)[T.KHM]	7.3300	2.472	2.966	0.003	2.486	12.174
C(country_code)[T.KOR]	2.4060	1.441	1.669	0.095	-0.419	5.231
C(country_code)[T.LKA]	4.2971	2.467	1.742	0.081	-0.537	9.132
C(country_code)[T.MDG]	2.7395	2.820	0.972	0.331	-2.787	8.266
C(country_code)[T.MDV]	5.3110	1.849	2.873	0.004	1.687	8.935
C(country_code)[T.MEX]	1.0291	1.721	0.598	0.550	-2.343	4.401
C(country_code)[T.MUS]	2.2594	1.673	1.350	0.177	-1.020	5.539

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	coef	std err	z	P> z	[0.025]	[0.975]
C(country_code)[T.NGA]	5.3906	2.632	2.048	0.041	0.231	10.550
C(country_code)[T.NPL]	4.7219	3.000	1.574	0.115	-1.157	10.601
C(country_code)[T.PAK]	4.3605	2.451	1.779	0.075	-0.443	9.164
C(country_code)[T.PER]	4.1879	2.214	1.892	0.058	-0.151	8.526
C(country_code)[T.PHL]	4.8190	2.278	2.116	0.034	0.355	9.283
C(country_code)[T.POL]	2.3690	1.382	1.714	0.086	-0.339	5.077
C(country_code)[T.ROU]	2.8715	1.849	1.553	0.120	-0.752	6.495
C(country_code)[T.RWA]	7.4624	2.924	2.552	0.011	1.731	13.194
C(country_code)[T.SAU]	2.7045	1.002	2.700	0.007	0.741	4.667
C(country_code)[T.TUN]	1.6675	1.719	0.970	0.332	-1.702	5.037
C(country_code)[T.URY]	1.0400	1.433	0.726	0.468	-1.768	3.848
C(country_code)[T.USA]	0.4410	1.034	0.426	0.670	-1.586	2.468
C(country_code)[T.UZB]	5.6797	1.653	3.437	0.001	2.441	8.919
C(country_code)[T.VNM]	5.8162	2.107	2.760	0.006	1.686	9.946
C(country_code)[T.ZAF]	1.0083	1.551	0.650	0.516	-2.032	4.048
C(Year)[T.2002]	-1.1480	0.597	-1.923	0.054	-2.318	0.022
C(Year)[T.2003]	0.0883	0.511	0.173	0.863	-0.913	1.089
C(Year)[T.2004]	0.8268	0.377	2.194	0.028	0.088	1.565
C(Year)[T.2005]	0.4528	0.546	0.830	0.407	-0.617	1.523
C(Year)[T.2006]	1.5075	0.534	2.823	0.005	0.461	2.554
C(Year)[T.2007]	1.3943	0.505	2.761	0.006	0.405	2.384
C(Year)[T.2008]	-0.2083	0.498	-0.418	0.676	-1.184	0.767
C(Year)[T.2009]	-3.5958	0.652	-5.517	0.000	-4.873	-2.318
C(Year)[T.2010]	0.6513	0.502	1.296	0.195	-0.333	1.636
C(Year)[T.2011]	0.2204	0.504	0.438	0.662	-0.767	1.207
C(Year)[T.2012]	-0.7287	0.465	-1.568	0.117	-1.639	0.182
C(Year)[T.2013]	-0.9063	0.429	-2.111	0.035	-1.747	-0.065
C(Year)[T.2014]	-0.5142	0.370	-1.391	0.164	-1.239	0.210
C(Year)[T.2015]	-1.1645	0.422	-2.757	0.006	-1.992	-0.337
C(Year)[T.2016]	-1.0264	0.470	-2.184	0.029	-1.948	-0.105
C(Year)[T.2017]	-0.7089	0.395	-1.793	0.073	-1.484	0.066
C(Year)[T.2018]	-0.5355	0.409	-1.308	0.191	-1.338	0.267
C(Year)[T.2019]	-1.2218	0.435	-2.808	0.005	-2.075	-0.369
C(Year)[T.2020]	-9.3816	0.872	-10.759	0.000	-11.091	-7.673
C(Year)[T.2021]	2.2876	0.842	2.718	0.007	0.638	3.937
C(Year)[T.2022]	-0.0907	0.539	-0.168	0.866	-1.147	0.966
C(Year)[T.2023]	-1.3368	0.669	-1.997	0.046	-2.648	-0.025
ENV_index	-0.6877	0.812	-0.847	0.397	-2.280	0.904
SOC_index	-0.0427	0.191	-0.224	0.823	-0.416	0.331
GOV_index	0.3860	0.356	1.085	0.278	-0.311	1.083

Omnibus:	385.644	Durbin-Watson:	2.037
Prob(Omnibus):	0.000	Jarque-Bera (JB):	33581.025
Skew:	-0.619	Prob(JB):	0.00
Kurtosis:	29.833	Cond. No.	107.

B Code Repository

GitHub Repository:

<https://github.com/Gabriel-pf193/sustainability-economic-performance>

Repository structure

The repository is organized as follows:

- **main.py**: entry point that runs the full pipeline in sequence (data preparation → country selection → fixed-effects regression → machine learning).
- **src/**: source code modules:
 - **PART_1_data_preparation.py**: merges the raw datasets and saves the full processed panel.
 - **PART_2_country_selection.py**: builds the 50-country sample and saves tables/figures related to sample selection.
 - **PART_3_FE_regression.py**: creates the regression dataset, estimates the fixed-effects model, and saves the regression table.
 - **PART_4_ML_models.py**: builds the ML dataset, trains the models, and saves evaluation outputs (learning curves/feature importance plots).
- **data/**:
 - **raw/**: downloaded datasets (original files).
 - **processed/**: intermediate datasets generated by the pipeline.
- **results/**: output files produced by the analysis (regression tables, model comparison tables, figures).
- **environment.yml**: conda environment specification for reproducibility.

Installation instructions

To recreate the environment and run the code:

```
conda env create -f environment.yml
conda activate esg-eco-project
python main.py
```

How to reproduce results

Running `python main.py` reproduces all main outputs of the project. Intermediate datasets are saved in `data/processed/`, and final tables/figures are saved in `results/`.

C AI Usage Declaration

This project was developed with the support of AI tools, in accordance with the course's AI Tools Policy.

AI tools used

- ChatGPT 5 (OpenAI)

How AI tools were used

ChatGPT was used as a learning and assisting resource throughout the course of the project. More precisely, it was used for:

- Clarifying concepts related to econometrics, machine learning, and data science
- Understanding and debugging Python errors
- Getting clarifications on the possibilities of Python libraries
- Receiving suggestions for code structure
- Receiving guidance on how to code specific steps of the project
- Improving clarity and wording of written sections in the report (e.g. explanations, paraphrasing)

All final decisions regarding the design, code, analysis, and interpretation of results were made by me.